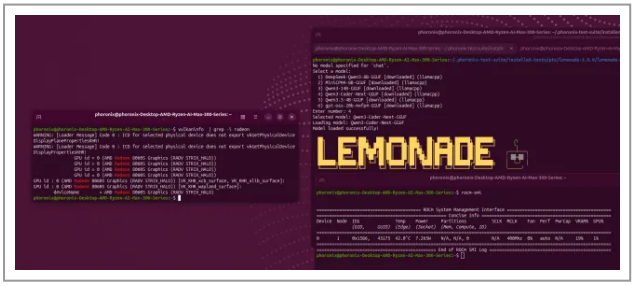
AMD ROCm vs. Vulkan Performance For Lemonade Local AI Server With Llama.cpp

Written by Michael Larabel in Display Drivers on 5 October 2026 at 09:50 AM EDT. **Page 1 of 3**. 15 Comments.

A Phoronix Premium reader recently inquired about seeing some new benchmarks of AI performance between Vulkan and AMD ROCm back-ends to see how the situation plays out today. Here are some fresh benchmarks on two separate systems with AMD Radeon graphics and using the Lemonade local AI server while exploring the difference between the ROCm and Vulkan Llama.cpp back-ends.



It's been nearly a year since last looking at the RADV Vulkan vs. AMD ROCm performance with Llama.cpp on AMD RDNA4. Going back then there were a number of models where using the Llama.cpp Vulkan back-end was faster than using AMD ROCm of the time.

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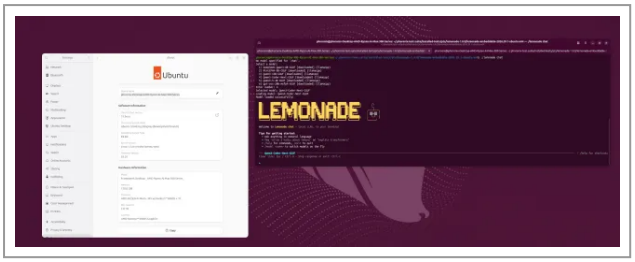
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With today's testing the Lemonade 2026.39.1 local AI server binary was used for testing for easy reproducibility with its built-in benchmarking support. Lemonade 2026.39.1 uses Llama.cpp b10825.

ROCm vs. Vulkan Lemonade Strix Halo

Phoronix.com	Phoronix Test Suite
AMD RYZEN AI MAX+ 395 @ 5.19GHz (16 Cores / 32 Threads)	Processor
Framework Desktop (AMD Ryzen AI Max 300) FRANMFCP06 (03.01 BIOS)	Motherboard
AMD Strix/Strix	Chipset
4 x 16GB LPDDR5-8000MT/s Micron MT62F4G32D8DV-023 WT	Memory
2000GB Western Digital WD_BLACK SN7100 2TB + 123GB SanDisk 3.2Gen1	Disk
AMD Radeon 8060S 64GB	Graphics
AMD Radeon HD Audio	Audio
DELL U5226KW	Monitor
Realtek RTL8126 5GbE + MEDIATEK MT7925	Network
AMD NPU	npu
Ubuntu 26.10	OS
7.3.0-rc4-phx-iommu-next (x86_64)	Kernel
GNOME Shell	Desktop
Wayland	Display Server
4.6 Mesa 26.2.2-1ubuntu1	OpenGL
GCC 15.3.0 + LLVM 21.1.8	Compiler
ext4	File-System
6144x2560	Screen Resolution

ROCm Vs. Vulkan Lemonade Strix Halo Benchmarks

- Transparent Huge Pages: madvise
- Scaling Governor: amd-pstate-epp performance (Boost: Enabled EPP: performance) - Platform Profile: performance - CPU Microcode: 0xb700037 - ACPI Profile: performance
- BAR1 / Visible vRAM Size: 65536 MB
- gather_data_sampling: Not affected + ghostwrite: Not affected + indirect_target_selection: Not affected + itlb_multihit: Not affected + l1tf: Not affected + mds: Not affected + meltdown: Not affected + mmio_stale_data: Not affected + old_microcode: Not affected + reg_file_data_sampling: Not affected + retbleed: Not affected + spec_rstack_overflow: Mitigation of IBPB on VMEXIT only + spec_store_bypass: Mitigation of SSB disabled via prctl + spectre_v1: Mitigation of usercopy/swapgs barriers and __user pointer sanitization + spectre_v2: Mitigation of Enhanced / Automatic IBRS; IBPB: conditional; STIBP: always-on; PBRSB-eIBRS: Not affected; BHI: Not affected + srbds: Not affected + tsa: Not affected + tsx_async_abort: Not affected + vmscape: Mitigation of IBPB on VMEXIT

System Logs

Two systems were used for this fresh ROCm vs. Vulkan testing, one a System76 Thelio Major workstation with the AMD Radeon AI PRO R9700 and the other the Framework Desktop with the AMD Ryzen AI Max+ 395 Strix Halo. Completely different hardware and slightly different software stacks to offer distinct looks at Vulkan vs. ROCm with Lemonade/Llama.cpp. Thanks to Phoronix Premium supporters that help make all of the daily Linux benchmarking possible at Phoronix 22+ years and going.

Performance On AMD Strix Halo

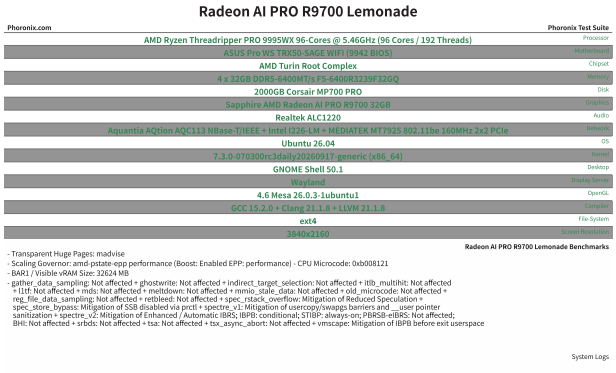
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